DAY 11

```python
rmse = np.sqrt(mse)

mae = mean_absolute_error(y_test, y_pred)

r2 = r2_score(y_test, y_pred)

print("MSE:", mse)

print("RMSE:", rmse)

print("MAE:", mae)

print("R2 Score:", r2)

#Step 7: Adjusted R²

n = X_test.shape[0]

p = X_test.shape[1]

adjusted_r2 = 1 - (1 - r2) * (n - 1) / (n - p - 1)

print("Adjusted R2:", adjusted_r2)
```

```
MSE: 0.555891598695242
RMSE: 0.7455813830127748
MAE: 0.5332001304956991
R2 Score: 0.5757877060324526
Adjusted R2: 0.5749637928613576
```

[4]:

```python
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

# Load dataset
df = pd.read_csv("student_data.csv")

# Features (X) and Target (y)
X = df[['Hours_Studied']]
y = df['Pass']

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
[4]: import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

# Load dataset
df = pd.read_csv("student_data.csv")

# Features (X) and Target (y)
X = df[['Hours_Studied']]
y = df['Pass']

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create model
model = LogisticRegression()

# Train model
model.fit(X_train, y_train)

# Predictions
y_pred = model.predict(X_test)

# Predict new student
new_student = pd.DataFrame({
    'Hours_Studied': [2]
})
prediction = model.predict(new_student)
print(y_pred)
print("Prediction (0=Fail, 1=Pass):", prediction[0])
```

```
[1 0]
Prediction (0=Fail, 1=Pass): 0
```

jupyter day11 Last Checkpoint: 2 minutes ago

File Edit View Run Kernel Settings Help

Code JupyterLab Python [conda env:base] *

```
[1]: ##Step 1: Import libraries

import numpy as np

from sklearn.datasets import fetch_california_housing

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score

#Step 2: Load dataset

data = fetch_california_housing()

X = data.data

y = data.target

#Step 3: Train-test split

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42

)

#Step 4: Train model

model = LinearRegression()

model.fit(X_train, y_train)

#Step 5: Predict

y_pred = model.predict(X_test)

#Step 6: Evaluate metrics

mse = mean_squared_error(y_test, y_pred)
```